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However, the direction of work changed markedly in the third e
ly 1993 to 2012. In this period, digital text became abundantly av
compelling direction was to develop algorithms that could achiev
language understanding over large amounts of natural text and th
istence of this text to help provide this ability. This led to a fundar
tation of the field around empirical machine learning models of N
tion that still dominates the field today. At the beginning of this pe
inant modus operandi was to get hold of a reasonable quantity of
those days, text collections were generally in the low tens of milli
and to extract some kind of model out of these data, largely by co
lar facts. For example, you might learn that the kinds of things pe
fairly evenly balanced between locations with people (like a *city*, *to*
metaphorical notions (like *imagination*, *attention*, or *essence*). But co
only go so far in providing language understanding devices, and
attempts to learn language structure from text collections were fa
ful.[6] This led most of the field to concentrate on constructing anno
resources, such as labeling the sense of words, instances of pers
names in texts, or the grammatical structure of sentences in treeb
by the use of supervised machine learning techniques to build mo
produce similar labels on new pieces of text at runtime.

The period from 2013 to present extended the empirical orientat
era, but the work has been enormously changed by the introductio
ing or artificial neural network methods. In this approach, words
are represented by a position in a (several hundred- or thousand
real-valued vector space, and similarities of meaning or syntax are
proximity in this space. From 2013 to 2018, deep learning provided
ful method for building performant models: it was easier to model
contexts, and models generalized better to words or phrases with
ings because they could exploit proximity in a vector space, rather t
on the identity of symbols (such as word form or part of speech)